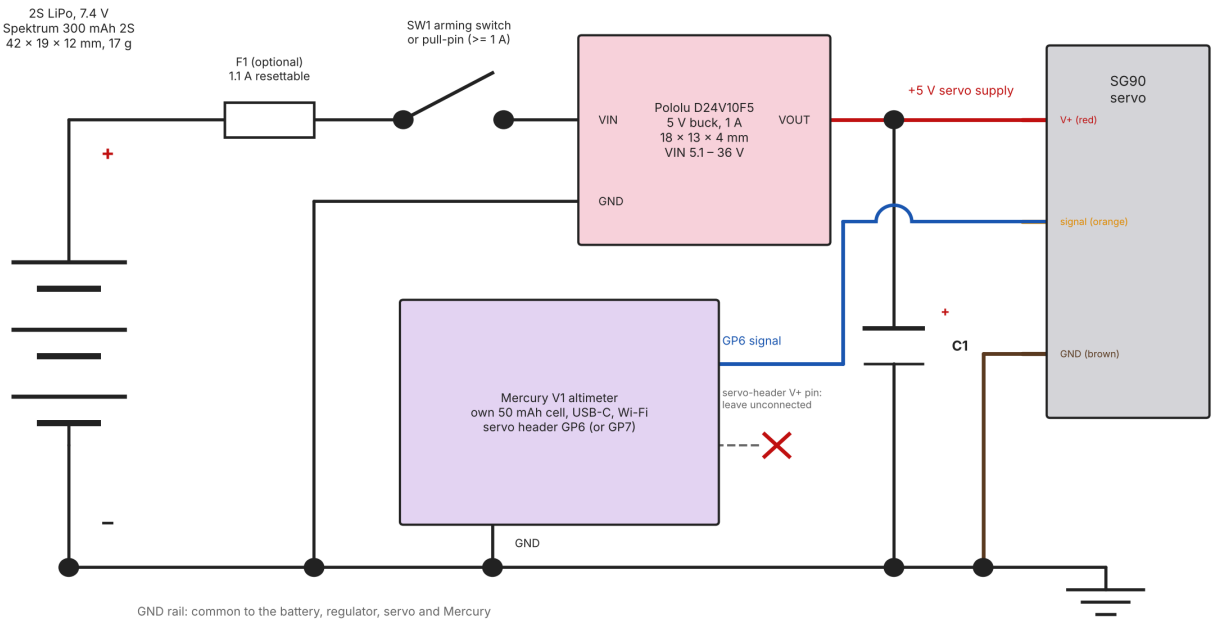
V4 wiring layout plan (after the design-review changes)

Where each wire runs inside and around the V4 model, drawn on sections of the rebuilt CAD. The electrical schematic is unchanged (see the power and wiring document); what is new here is the routing, and the routing turned out to need changes to the CAD. The model is the V4 studio after the review changes (battery pocket for the Spektrum 300 mAh 2S, strap channels, wider bearing spokes).

Main finding: as modelled, the servo cable has **no route** to the bay, and the cable has no way into the sled pockets either. The earlier idea of running the servo wires down the hollow rotor does not work: the horn closes the top of the bore (0.2–0.5 mm gaps). Four small CAD changes (P1–P4 below) are proposed; **none is applied yet**, because P1 and P2 depend on where the Mercury's servo header is (I could not read its manual), and you should pick between options.

1 Harness (unchanged)

V4 airbrake: servo power and signal (wiring schematic)



C1 = 470 - 1000 uF, 10 V, low-ESR, wired right at the servo pins. The regulator powers the servo; the Mercury only sends the position signal.

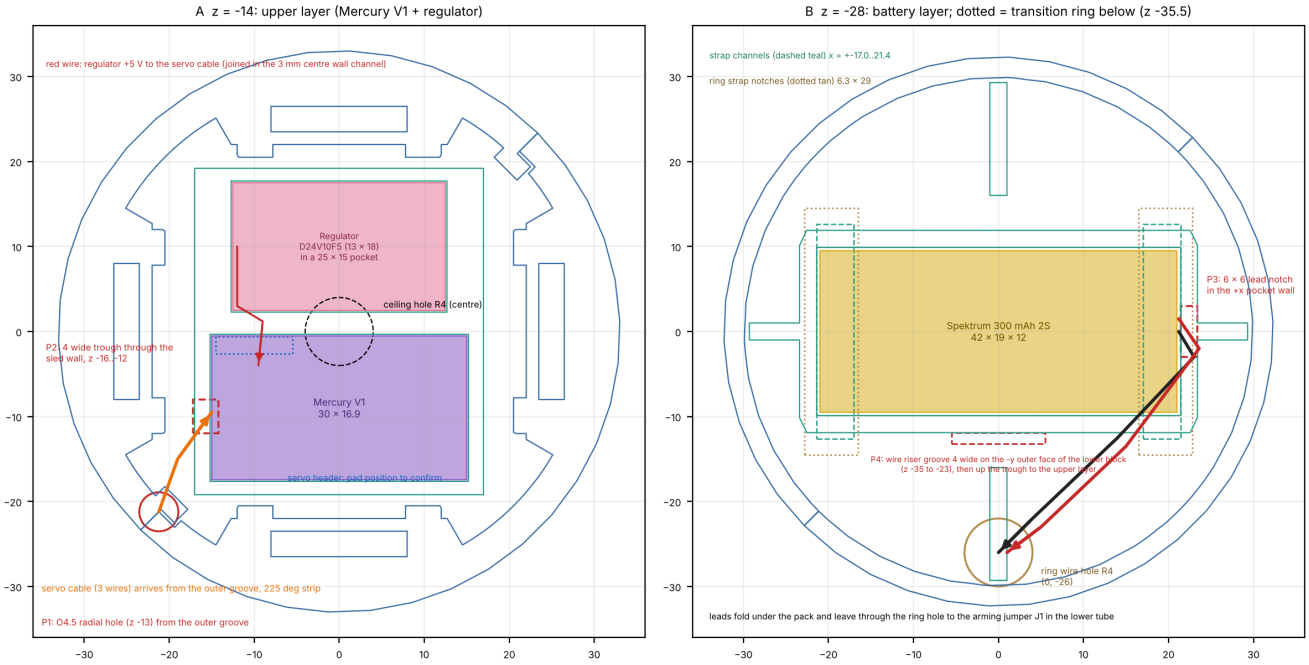
Ground is shared. Never join the servo red wire to a Mercury pin; confirm the Mercury servo-header pin order in its manual before wiring.

Reference schematic (battery label now Spektrum 300 mAh 2S).

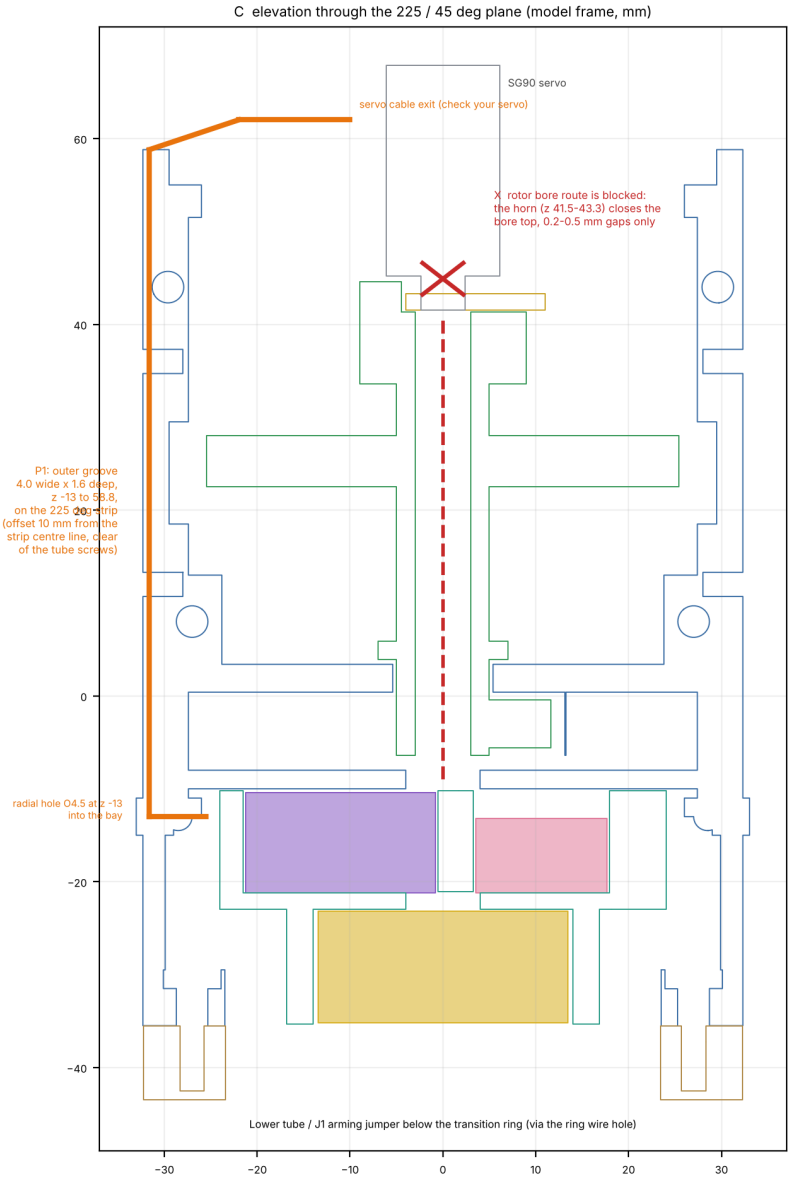
Wire	Function	From → to	Wire	Length (est.)
W1	Battery + through the arming jumper	pack plug → polyfuse → J1 (lower tube) → regulator VIN	24 AWG red	60 mm + 110 mm
W2	Battery –	pack plug → regulator GND (the ground rail)	24 AWG black	105 mm
W3	+5 V	regulator VOUT → C1+ → servo red (inside the servo cable)	24 AWG red	40 mm + cable
W4	Ground rail	regulator GND → C1– → Mercury GND → servo brown	24 AWG black	40 mm + cable
W5	Servo cable (3 cores)	servo → outer groove (P1) → radial hole → bay → splice point	servo's own lead	about 150 mm
W6	Signal	Mercury GP6 (or GP7) servo-header signal pin → servo orange	28 AWG	25 mm
J1	Arming jumper	remove-before-flight 2-pin plug in the lower tube, just below the ring wire hole	JST-PH or header	—

Lengths are read off the routes drawn below (±15 mm) and do not include slack: add about 20 mm to each. The servo's own lead is usually 200–250 mm (seller dependent): enough if you cut it at the bay end. Cap C1 (470–1000 µF, 6.3 × 11 mm) stands beside the regulator in its pocket (the pocket is 11.3 mm deep).

2 Layout drawings



Plan sections of the rebuilt model with the routes (orange: servo cable, red: +5 V and battery +, black: battery –; dashed red: proposed openings).



Elevation through the 45°/225° plane. Orange: proposed servo-cable route (P1). Red dashed with a cross: the rotor-bore route that does not work.

3 What is in the way, and the proposed changes

Id	Problem found while drawing the routes	Proposed change (not applied)	Touches
P1	Servo cable: the horn blocks the rotor bore; the housing wall around the strips is too thin for a hole; the 0.7 mm gap under the upper tube is too small.	Cut an outer groove 4.0 wide × 1.6 deep in the strip at azimuth 225°, offset 10 mm from its centre line (clear of the tube screws), from z –13 to the housing top (58.8). Add a radial O4.5 hole at z –13 into the bay. Remaining wall 1.2 mm; strip carries no load except the tube screws.	housing halves; check cam peak 28.6 vs wall 29.5 (clear)
P2	The ceiling hole and the pocket tops are only 0.2 mm below the ceiling, so no wire can enter the Mercury or regulator pockets.	Cut a 4 wide trough through the sled wall at the cable side (x –17.2…–14.2, z –16…–12) and a 3 mm channel between the two pockets; position finalised once the Mercury header position is known.	sled
P3	Battery leads leave the pack end, where the pocket wall is only 0.4 mm away.	Cut a lead notch 6 wide × 6 tall in the +x pocket end wall (y ±3, z –34…–28), open to the bottom, so the leads fold under the pack toward the ring hole.	sled
P4	The two layers connect only through the R4 floor hole, which is closed below by the battery (0.2 mm).	Cut a 4 wide riser groove, 1.5 deep, on the –y outer face of the lower block (z –35 to –23) so battery+ and – reach the upper layer outside the pack.	sled
P5	Static pressure (existing open item).	Three O3 vent holes through the lower tube and the housing wall at z –16, azimuths 0°, 120°, 240° (Mercury level).	housing, tube

4 Harness assembly order

- Bench-test first: servo, regulator and Mercury on a table with a 5 V supply; confirm the servo moves from the Mercury signal alone (power only from the regulator).
- Crimp or solder the servo lead (P1 route), and pull the servo’s red wire out of the Mercury-side connector so only signal and ground reach the Mercury.
- Fit the servo to the ledges (M2×6, O1.8 pilots), feed its cable down the outer groove and through the radial hole into the bay.
- Fill the sled: regulator and C1 in their pocket, Mercury in its pocket, wires through the trough; battery in the bottom pocket with its leads folded out through the notch; strap it (two ties or one hook-and-loop strap).
- Run battery + and – down through the ring wire hole to J1 in the lower tube and back up the riser groove; leave J1 open (disarmed).
- Slide the sled up, fit the transition ring, check that no wire is pinched between the sled and the ring (the ring has strap notches for exactly this); only then close the jumper.

5 Open items for you

- The **Mercury servo header**: which pad is GP6/GP7, pin order, and signal voltage (3.3 V into the SG90 at 5 V usually works; bench-test it). P2 is placed once the header position is known.
- Where the **servo cable leaves your servo** (the drawing assumes the top end when the servo is flipped), and its length.
- Do you want **P1–P5** applied as drawn, or a different cable route (for example a slot in the servo ledge and a groove on the inside wall)?
- The Spektrum pack’s **plug type** (PH 2.0 listed, check) and the length of its leads.